

Fuzzy Logic in Recommender Systems

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Abstract A recommender system studies the past behaviour of a user and recommends relevant and accurate items for the user from a large pool of information. For user ‘ u ’ a recommender system filters relevant and accurate information by finding items which are similar to the target items (items searched by user) and by finding (other) similar users which co-relate to the user ‘ u ’ interests and needs. For exhibiting this filtration, a recommender system uses the features of the items and maintains user profile which contains his past purchases, his buying pattern. These features and user profile imprecise, uncertainty and vague thus should be analysed carefully for optimal prediction. Fuzzy logic has been extensively used in the design of a recommender system to handle the uncertainty, impreciseness and vagueness in item features and user’s behaviour. In this chapter, the use of fuzzy logic in recommender system as well as the analytical framework for analysis of the design of a recommendation system is discussed. It also presents the analysis of the growth of fuzzy logic in recommender system and its applications.

Keywords Collaborative filtering • Content based filtering • Dimensions of recommender system • Fuzzy logic • Recommender system

1 Introduction

Any business which involves electronic transaction is popularly known as E-commerce. As the World Wide Web (WWW) is growing day by day, the buying pattern of the people has drifted from brick and mortar stores to online buying [1]. So, data over the internet has increased manifolds.

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Nowdays, online buying has encouraged companies to work out more and more options for the customers to select from. They are customizing the products and increasing the feature set to suit various needs of multiple customers. But all the features of the products may not be useful for every customer and the customer may get confused due to the information overloading of the various features. Subsequently, the customers are interested to analyse all the information of the related products before actually buying the most suitable product for them. *Recommender system (RS)* is a good solution to efficiently deal with this information overload [2].

A recommender system is an automated tool which suggests appropriate items to a user [3]. A recommender system consists of a set of users ($u_1 \dots u_i$) and a set of items ($i_1 \dots i_k$). According to the definition, a user in a recommender system is an entity which uses a recommender system [4]. An item can be a tangible or digital object which is suggested to a user through text message, e-mail [4]. All items have a set of features. For example, an item can refer to an electronic product, a car, a movie or a song which individually have some features of their own. For instance, a car has features like mileage, boot space, leg room, exteriors, interiors, turning radius and so on. These features define an item.

Further, e-commerce uses recommender system to suggest products/items to customers of their interest. The items which are suggested to the customer are based on various factors, like- previous buying behaviour of the customer (historical data), personal information of the customer, most rated items similar to customer's interest. Thus, a recommender system has the ability to give personalized recommendation [4] to the user. The advantage of such a system is that it saves valuable time of a customer and increases the loyalty of the company as well [5]. Further, this personalization in recommendation, broadly, comes from two categories - *explicit* or *implicit user behaviour*. For instance, explicit user preference is when a user u_i gives a rating r_j to an item; and implicit preference is when the system follows the clickstream information of the user. A reference model [6] to depict these preferences is given in Fig. 1.

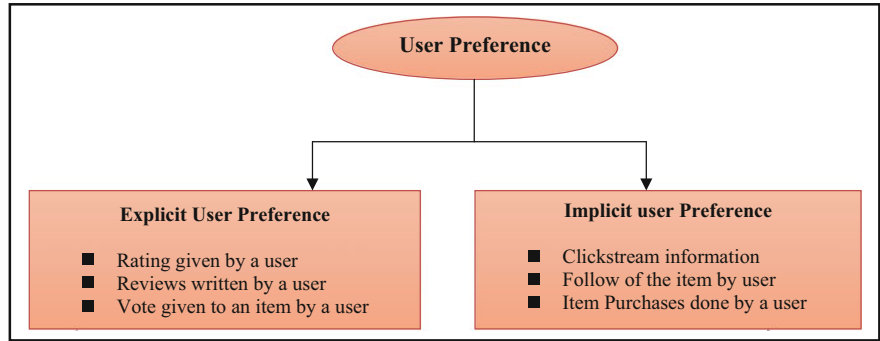


Fig. 1 Reference model of user preference

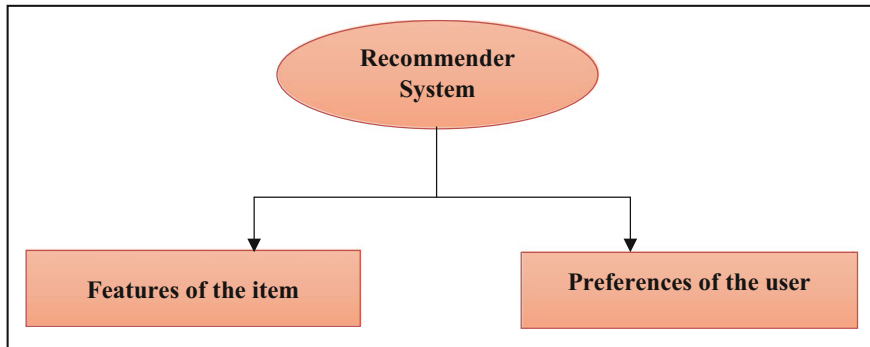


Fig. 2 Factors affecting a recommender system

A Recommender System works on the features of the item as well as on the preferences of the user, as given in Fig. 2. The performance/efficiency of a recommender system depends on these factors and how they are represented. Both these factors play a vital role in demonstrating the level of personalization to a user.

But these factors are subjective and imprecise. This poses a major challenge for recommender system to handle the uncertainty [7, 8]. It is also interesting to note that these uncertainties have an extent (low to high) to which the items have features. For example, a car can have a feature, such as *interiors* as very good, good, not so good, bad, and awful.

To handle this uncertainty, use of fuzzy logic in recommender system is highly desirable. This not only helps in modelling the vagueness in a logical manner but also gives a consistent justification and heuristic rules for designing a recommender system [9]. In literature, fuzzy logic has been used to design a recommender system [10, 11]. With the application of fuzzy logic [12, 13], the recommendation has seen the following major changes:

- a. the fuzzy rules help to model the vagueness in user input in an efficient manner with the help of linguistic variables (“like”/“dislike”/“awful”/“good”).
- b. use of fuzzy logic with other techniques like clustering improves the determination of similar users, by finding the fuzzy similarity degree between two users,
- c. the closeness between two fuzzy ratings (user input) over a particular product has been understood in an easy way with the help of fuzzy rules,
- d. handling of user demographic data is better with fuzzy rules.

To understand the use of fuzzy logic in recommender system let us consider a recommender system which recommends an automotive to a user. To generate the recommendations which are most suitable to a user; the system inputs certain priorities on various parameters which the user can set. For instance, his budget, average kilometre coverage, fuel efficiency, styling, comfort, service, space, infotainment and so on. If a user is asked to set his parameter priorities on a scale of 1 (least important) to 5 (most important), then instead of giving a crisp value to these

parameters; he gives a degree of his likeness i.e., for styling he may set the value as 3–5. Human reasoning works better on range of values as it better captures the vagueness in deciding a particular parameter. Taking this as an advantage, the recommender system can use fuzzy rules to map user ratings, demographic data (such as age, income, and profession), and similarity of two users more efficiently.

This chapter aims at providing a support to the use and need of fuzzy logic in recommender system, and to motivate the readers to find new applications in various dimensions of the recommender system.

In this chapter, Sect. 2 discusses the idea behind the origin of recommendation. Section 3 explains the analytical framework of a recommender system. The understanding of the framework helps in analysing the future uses of recommender system with other techniques like fuzzy logic, genetic algorithms. In Sect. 4, a brief analysis of fuzzy logic in recommender system is presented. It gives an overview of the growth of fuzzy logic based recommender system in literature since 2003. A conclusive remark of this chapter is given in Sect. 5.

2 Idea of Recommendation—Social Information Reuse

Information reuse has been a key area of research. Social navigation is a branch of information reuse which tries to understand the interaction among people. Social navigation (now social network) refers to the idea of following the footsteps of others in order to find out what one wants [6].

The use of social navigation can be understood from a story of cavemen who used to hunt for food. Once one of them found a bright red apple and died after eating. The fellow cavemen *used this information* and did not eat that fruit. Humans have a tendency to learn from other’s experience/actions [6]. With today’s digital age, as the data over the internet rose to a level, it gave rise to two technologies—*information filtering and information retrieval*. The basic differences between these techniques [14] is summarized in Table 1. The difference between information

Table 1 Comparison of information filtering with information retrieval techniques based on different aspects

Aspect	Information filtering	Information retrieval
Users of the system	Use by a person/persons with long term interests	Use by a single person with one time goal
Information representation	User profiles with specifications	In the form of queries
Basic functioning	Distribution of information to groups or individuals	Collection and organization of information
Database type	Dynamic database	Static database
User related issue	Generally, for undefined group of people seeking advice with varied domains e.g. entertainment, product service etc.	For user’s belonging to well defined groups with specific domains

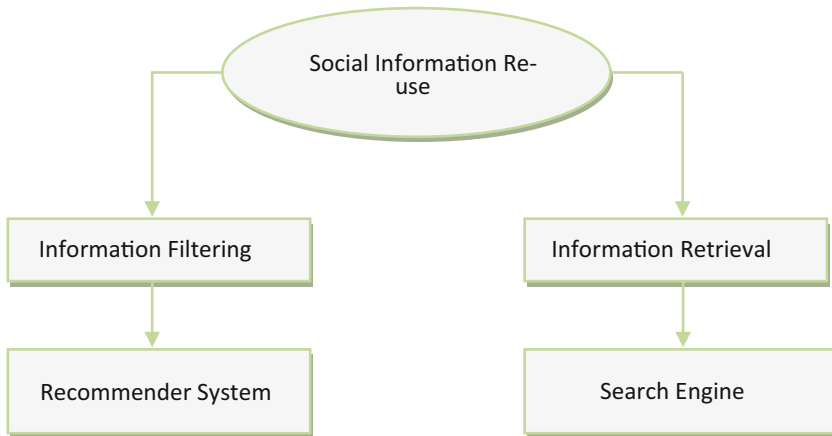


Fig. 3 Recommender system as an information filtering system

filtering and retrieval is important. It clearly defines the characteristics of a recommender system (*information filtering*) and a search engine (*information retrieval*). Although both information filtering and retrieval are seen as two sides of the same coin [14]. But it is always useful to understand the difference between the two.

From Fig. 3, it can be understood that a recommender system is an information filtering system which generates intelligent agents for searching remote, heterogeneous databases [15]. For instance, Netflix [16] recommends movies to the user based on his interest by going through a large set of databases in order to predict whether a particular item will be useful for a user or not based on user/item profile. Information retrieval systems are those which provide documents to users which satisfy his/her need for information [14]. System which are popularly used for information retrieval are search engines like Google.

A recommender system is considered as an information filtering system because it dynamically filters out vital information from large databases based on user's preference, behaviour or interest [17]. Sometimes there is a confusion regarding the differences between recommendation and prediction. In prediction, the focus is on estimating the degree to which an item will be liked by a user and it is often concerned with the browsing of specific products. A recommender system helps to suggest items/products most suitable to a user or items that he will like [18]. A recommender system has the ability to predict the items/products to a user based on his interest. The recommender system can be classified as—Content Based [19], Collaborative Filtering Based [20] or hybrid recommender system. A detailed classification of recommender system is given in the subsequent sections.

3 Analytical Framework of a Recommender System

A recommender system is an information filtering system which collects the opinion and experience data of users to find relevant data items *for a purpose* [21]. The role of a recommender system is to compute the recommendations and present them in a useful way to the user [19]. This section, explains the analytical framework consisting of various dimensions which are useful in understanding the basic design and functioning of a recommender system [6]. These dimensions are: *Domain of Recommendation*; *Purpose of Recommendation*; *Recommendation Context*; *Level of Personalization*; *Trustworthiness of Data*; *Interfaces and Recommendation Algorithms*. Figure 4 shows a diagrammatic view of these dimensions [6].

3.1 Domain of Recommendation

Domain of recommendation is concerned with the items to be recommended to the user. The item can be a single or a bundle or sequence of items? For example, the item can be a music piece/list of songs, news articles, or electronic product. It works on the principle of *interestingness of an item* with other items. It means, instead of recommending a particular item X, a bundle of items similar to X are recommended. Domain also deals with the ordering of the items in the recommendation

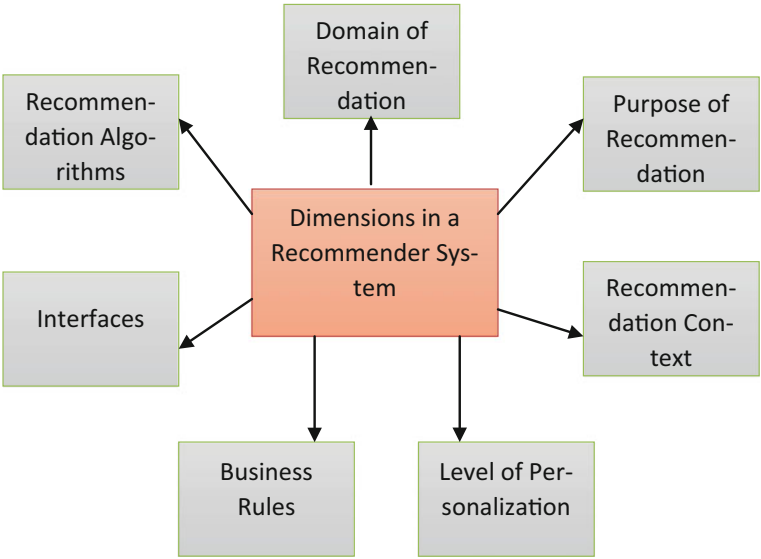


Fig. 4 Analytical framework of a recommender system

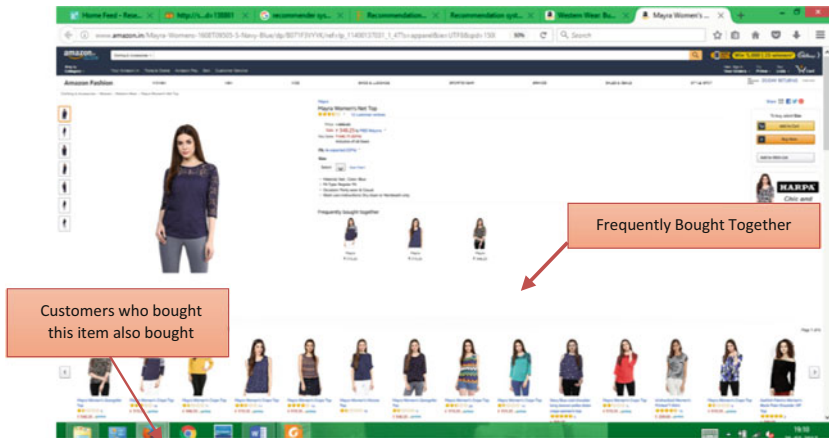


Fig. 5 A page from women fashion (western wear) from Amazon.in

list [23]. For instance, recommending a list of songs to a user, the ordering of the playlist plays an important role. It is usually populated with songs which the user has heard before. So the domain knowledge of what the user has liked before is important. Another example is when a user thinks of purchasing an item, the recommender system, by knowing the user’s interest through selected item, can be given other options to choose from. For instance, Fig. 5 represents a page from *Amazon.in > Women Wear > Western Wear*. Here Amazon’s recommender system outputs two ways of recommendation: first, “*frequently bought together*” and second, “*customers who bought this item also bought*”. This gives the user a wider array of selected options from a large set of western wear. Also, the user gets the recommended list with *almost* the same price range (Fig. 5).

3.2 Purpose of Recommendation

In the entire process of recommendation, the aim is to output the most desirable recommendations to the user based on his interest. In purpose of recommendation, the idea is to make aware the community/educate for a particular facility. For example, TripAdvisor (a recommender system and a travel review site) educates the customer to book the travel to a place which is top ranked or give a recommendation score to all the hotels which are similar to the customer’s query. It never compels a customer to book hotels/restaurants/flights. This system is efficient and powerful enough to educate the customer beyond recommendations [6].

3.3 *Recommendation Context*

The recommendation context is a spring board for a recommender system. It seeks two important aspects of a recommender system. First, the *level of attention* a user has while the system recommends items to him and second, *degree of interruption* a user is willing to take. Now let us understand these aspects with the help of examples. If a user is listening to a song, then, he will dislike any kind of interruption, be it a list of further recommended songs. As he just wants to enjoy his favourite track. Also, the context of recommendation has to understand the level of attention the user has. For example, for a movie freak, if the system recommends some good movies (the definition of “good” is handled by the recommendation algorithms; explained later) based on the situation that person is in, like, he is alone or with his friends or with family. The recommended movies will change in every context (if not all).

3.4 *Level of Personalization*

A recommender system personalizes the information to increase the interest of the user. This personalization can be based on the age or gender or profession. For example, an online clothing store might recommend different clothing options based on different age groups like 18–25, 25–35 years and 35 above. It is observed that the level of personalization affects the user’s interest in a recommendation system [6].

3.5 *Business Rules*

Apart from understanding the context of recommendation, it is important to know whether a recommender system is trustworthy or not. The trust on a system is governed by rules which run the purchasing properties of a customer. *For instance*, a customer should be made assured that the product he is buying is the best and has the lowest price as compared to other recommendation systems. Now the company has to decide on two factors—first, *trustworthiness of data* and second, its *profit making strategy*. It is also vital to know about the biases, a company has [6]. The biases are often called as business rules. These business rules are used to increase the purchase of products by a user. This can be achieved by showing complementary products and the help in finding related products also [23]. *For instance*, it is very uncommon to see, that a shoe seller would admit that he does not sell the shoe that the user is looking for. Recommendations are governed by business rules, after all, these are generating revenue for the company. So, in short it can be said

that a recommender system should be able to handle the issues of transparency and the trust worthiness of the users, who are giving ratings for a particular item.

3.6 Interfaces

In this dimension, the idea is to understand, what the final output of a recommendation system is. It can be in the form of *prediction*, *filtration*. Prediction gives specific scores for an item. Filtration gives the best result from a given search list. A recommender system can be an agent which gives helpful recommendation; it can be give a dialog interface where the user can show his like or dislike for a recommendation; or it can act as a search tool. It is also concerned with the type of input from the user—either implicit or explicit. This has been explained in Sect. 1. For a thorough discussion on interfaces readers can have a look at [19, 22].

3.7 Recommendation Algorithms

The recommendation algorithms can broadly be classified as: *Non-personalized summary statistics, content based filtering, collaborative filtering and hybrid*. These algorithmic models are explained in this section. Figure 6 depicts the broad classification of recommender system.

a. Content Based Filtering

Summary statistics are similar to calculating the average number of users who have rated a particular item. For example, how good a movie is? does not require any personalization. It can be computed without the use of a model. The advantage of using summary statistics is when many people rate a particular item/product the impact of each rating is low; this helps in dampening any individual outlier rating. But there are issues which need to be addressed while using *non-personalized*

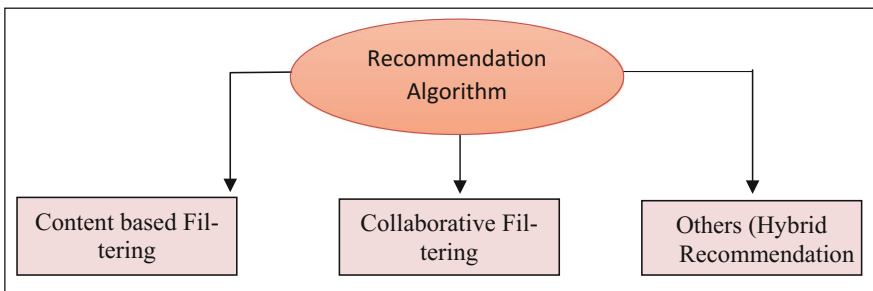


Fig. 6 Types of recommendation algorithms

summary statistics, such as, people with *dissimilar taste* when rate a particular item, or people with *strong opinions* moves the average rating to a higher value or sometimes the averages can be *biased* (when a user self-rate a movie/product/any other item). This method lacks the consideration of context, which is important for recommendation. For example, if someone at *Subway* (a global restaurant chain) wants a recommendation for a sauce, average statistics will say a tomato ketchup; but if he asks for the most likable sauce associated with a veg burger, then this models fails to find this association. Also, Non-personalized systems give similar recommendations to all users without considering individual tastes/preferences. This leads to designing recommender system where Product Association Recommendations are considered. The content based systems consider *Product Association Recommendations* (PAR). It helps to find other products that go well together with the target product. The co-purchasing of a product is a major consideration which is used to satisfy business rules. The rules have certain goals [6, 23]. The aim is to increase the number of products a user buys. It is also to show user complementary products to help him find the most related products. For example, Fig. 7 shows a webpage from *ebay.in* (online commerce company), where a user wishes to purchase I-phone (target product). The highlighted “similar items” in Fig. 5 show the content similar to the target product.

A major limitation of this method is the inability of the system to show the user “the most popular” products. It also gives the same recommendation to every user and does not look at individual differences. In scenarios where the user is visiting just once, it is very hard to build a profile. Similarly, if the user’s taste change rapidly, this recommendation may not be very useful as the recommended list of items will be out of date.

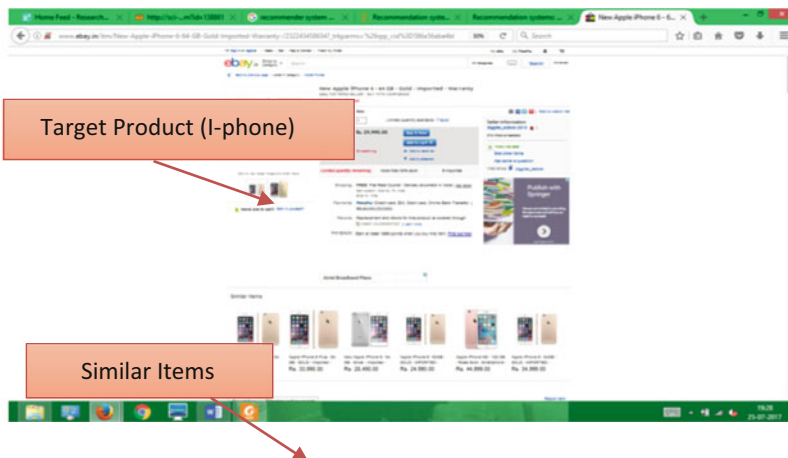


Fig. 7 A webpage from *ebay.in* for demonstrating content based recommendation

b. Collaborative Filtering

To overcome the limitation of content based filtering, a *user-user* and an *item-item* collaborative filtering method is used. Collaborative filtering (CF)-based recommendation techniques enable the users to choose on the basis of opinions of other people who have similar interests. The collaborative filtering is of two types—*user-user* and *item-item*. In user-user filtering, the recommendations are based on users who share similar interests. In item-item, the user is flooded with recommendations for the items that are loved by other users in the past who have shared similar interests earlier. Similarity measures are used to find the likeliness of two users. The similarity measures used are: cosine similarity, Pearson co-relation or adjusted cosine. These measures can be studied in [18, 20]. For using these measures, only those users who have rated both the items are considered. This poses a bottleneck for new user.

In collaborative filtering the prediction for a user ‘*i*’ in personalized and non-personalized scenarios is given by Eq. 1.

$$P(a, i) = \sum_{u=1}^n \frac{r(u, i)}{n} \quad (1)$$

where, ‘*p*’ is the prediction for user ‘*a*’ for item ‘*i*’ and the prediction is done for user ‘*a*’ based on other users ‘*n*’, $r(u, i)$ is the rating by user ‘*u*’ in item ‘*i*’.

Now there are two considerations, first, if item ‘*i*’ is not liked by everyone and they have different opinions about item ‘*i*’; second, if the ratings used for item ‘*i*’ are not on the same scale. These situations have to be handled differently.

In the first situation, for predicting item ‘*i*’ to user ‘*a*’, if the item is not drawn evenly from all users then, we use a personalized weight for every user and sum it over all. This is given by Eq. 2.

$$P(a, i) = \frac{\sum_{u=1}^n r(u, i) * w(a, u)}{\sum_{u=1}^n w(a, u)} \quad (2)$$

where, $r(u, i)$ is the rating by user ‘*u*’ for item ‘*i*’, and $w(a, u)$ is the weighted agreement between user ‘*a*’ and user ‘*u*’. It is given by the similarity between user ‘*a*’ and user ‘*u*’.

The second situation, where ratings are not on the same scale, the difference of ratings of user ‘*u*’ for item ‘*i*’ and the average rating for user ‘*u*’ is taken as given by Eq. 3.

$$P(a, i) = \sum_{u=1}^n r(u, i) - r(u)/n + r(a) \quad (3)$$

where $r(u, i)$ is the user item pair and $r(a)$ gives the deviation which is added to the average rating of user ‘*a*’.

Collaborative filtering assumes that the past agreements of the users are helpful in predicting their future agreements. It also considers that the individual tastes of the users are stable and do not change over time and are in sync with one another. So, in collaborative filtering the system has to carefully work in the domain of agreement. Despite this assumption, there are situations when people agree on one domain and not on another. For instance, some people are happy having a discussion on politics but they might not necessarily agree on technical discussions. For such cases, the prediction for item ' i ' are given by Eqs. 2 and 3.

c. Hybrid Approaches

Both Content based and Collaborative systems have some limitations. Content based systems give recommendation on content profile of item. In some cases, it is difficult to generate content profile for every item. A content based system is unable to select items of interest to user, if the user has not given any evidence of it (*over specialization*). The content based systems are unable to distinguish between subjective information (*subjective domain*).

In collaborative filtering systems these limitations are overcome by adding user profile which can handle the *over specialization* problem and the *subjective domain* problem. But collaborative systems have certain problems of their own. It suffers from the early rater problem, gray sheep and data sparsity problem. A collaborative filtering system cannot make predictions for new items which have not been rated by any users. Similarly it is difficult to find accurate predictions for new users who have rated very few items. As discussed previously in collaborative filtering systems, the base assumption is that the tastes of the users do not change over time. But in reality, all users cannot have overlapping characteristics. Everyone cannot agree on every domain. So it is challenging to find accurate recommendations for gray sheep (user or group of users who do not have consistent agreement). Also, predicting those items which are not rated by enough number of users is also a difficult task because the data for that particular item is sparse [24]. Hybrid recommendation systems are the system which use the strengths of both the systems to efficiently handle recommendations [25–28].

4 Fuzzy Logic in Recommender System

Fuzzy logic [12, 13] gives a wide spectrum of methods to analyse the uncertainty in data. It is very helpful in handling data with imprecise information and gradualness of user preference [9]. Over the years, use of fuzzy logic to various areas has been studied. In this section, an analysis of the growth of fuzzy logic based recommender system is presented. It also discusses the applications where fuzzy logic based recommender system has been used.

4.1 Fuzzy Logic Based Recommender System: An Analysis

This sub-section presents the statistical analysis of fuzzy logic in recommender system. The data is collected and analysed from *web of science* (WoS) database to identify the growth of fuzzy logic based recommender systems. This analysis aims to provide the reader with the growth of fuzzy logic in recommender system, its major areas of applications from the time of its inception. It is interesting to note that fuzzy logic in recommender system has been extensively used in Computer Science as compared to other areas. A summary statistics has been shown in Table 2 (correlated from WoS). The research articles, since 2003, where fuzzy logic has been studied with recommender system is analysed.

Within Computer Science, the use of fuzzy logic has been studied in various areas (not limited to) like automatic group recommendations [29], knowledge based recommendations [30], software project management [31], E-Learning [32], multi-criteria collaborative filtering [33], trust model [34] and so on. The growth of these areas has seen various methodologies which are used for designing an application centric recommender system.

Recommender System based on fuzzy logic has been in use since, 2008. The year wise growth is depicted in Fig. 8. The growth of research in fuzzy logic based recommender system suggests a gradual increase from 2003 to 2012. It is the time when the potential of these systems was realized. Then a sudden increase in the research interest in 2013 is seen. It is during this time, that fuzzy logic based recommender system was applied to various applications like consensus ranking [35], item and trust-aware collaborative filtering [36], co-relation based similarity [37], competence recommendation [38], situation-aware collaborative filtering [39], tourism system [26], telecom products [27], personalized e-services [40, 41], stock market [42], movie recommendation [43].

Table 2 Use of fuzzy logic based recommender system in various research areas

S.No	Research area	Percentage
1	Computer science	92.31
2	Engineering	25.00
3	Operations research management science	15.39
4	Mathematics	6.73
5	Business economics	3.85
6	Telecommunications	3.85
7	Automation control systems	1.92
8	Education educational research	1.92
9	Information science library science	1.92

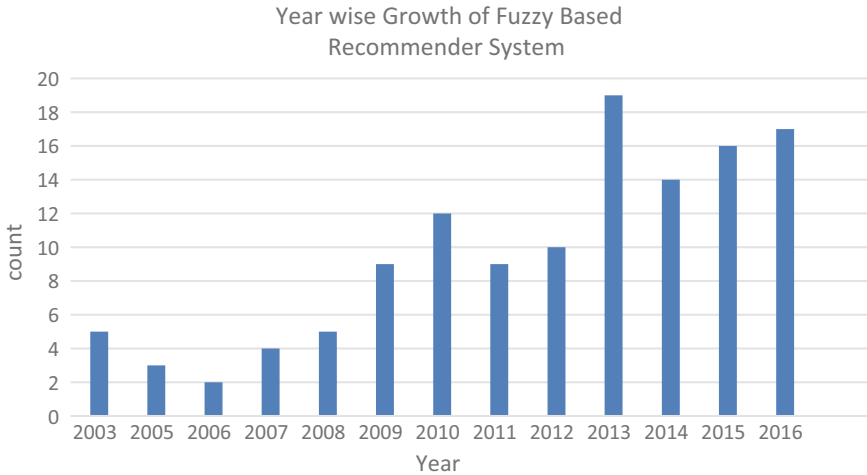


Fig. 8 Year wise growth of research in fuzzy logic based recommender system

4.2 Applications of Fuzzy Logic in Recommender Systems

As seen in Sect. 4.1, the growth of fuzzy logic based recommender system has been on an increase since its inception. In this section, the rise in the growth has been analysed with respect to the work done in fuzzy logic based recommendation since 2003. Yager in his seminal work [9], defined a fuzzy set A in X is defined by a membership function, which is defined as: $\mu_A(x): x \in X \rightarrow [0,1]$ where X is the domain space. So set A is characterized by $A = \{x, \mu_A(x), x \in X\}$. The membership function can have different interpretations based on the applications. For instance, it can be the degree of similarity or user preference [41, 42].

Since the idea of fuzzy logic based recommender system in 2003, this area of research has seen tremendous changes in the applications where the use of recommender system can be very well understood. The use of fuzzy sets to analyse recommender system design was a breakthrough to the world of recommendation [9]. The uncertainty in recommending accurate items/products to users has drawn attention. The uncertainty in representation of objects was studied. This representation implies that whether a system is content or collaborative, both need a vector of ratings for finding out the correct user interest. The initial methodology was developed for reclusive systems. In 2005, the use of fuzzy logic was studied with data mining techniques for recommender system. In [10] dynamic fuzzy cluster was proposed which used dynamic degree of membership for a user preference. In [11] fuzzy logic based recommender system monitored two considerations. These were—*consumers past purchases* and his *specific requirements*. This system aimed at guiding the consumers in electronic products. The initial work in 2006 used fuzzy sets and related reasoning techniques to model the user preferences and analysed case based recommendation system [44]. Following the same year, fuzzy

association rules and multiple similarity measures (FARAMS) were used to design a recommender system using collaborative filtering [45]. The technique was simple and effective but suffered limitations with de-fuzzification process.

By this time, fuzzy logic was realised as a potential breakthrough in dealing with uncertainty of data in recommender system. Researchers focused on the granularity of data. The granularity in creating user profiles and for the items which remain unsold due to non-repetitiveness of the stock. In [46] a fuzzy hierarchical clustering algorithm was used to cluster similar users. The users within a cluster shared common characteristics. The algorithm worked on log data. This method was efficient than previous modified versions of fuzzy c-means. Another interesting property for collaborative filtering system was explored in [47] which considered irregularities in rating of item which could not be sold “repetitively”. This approach used graded membership for domains of similar users to eliminate the possibility of *one-and-only items*.

After analysing fuzzy logic in recommender system, researchers realised that there are variable entities such as user demographics in recommender system which can be represented with a graph. Then, probabilistic graphical models were used with recommender system. In [48, 49], a combination of Bayesian networks and fuzzy set theory is used to recommend new items to the users based on the past voting patterns of users with similar taste. Along the statistical models, researchers began to use the modelling of user and rating with fuzzy linguistics. In [50], the information in technology transfer office [25] is used to design a recommender system which uses fuzzy linguistic modelling. It was studied extensively for managing subjectivity [51], for acquisition of user profiles [52], managing digital libraries [53].

Another important aspect of recommender system was dealt with fuzzy sets. Fuzzy sets were used to design a system which could adapt to user’s changing behaviour. This changing behaviour (with time) brings dynamicity in the model. In [54] fuzzy set theoretic method were used to handle the vagueness and imprecision in data. It extends the fuzzy modelling proposed in [9]. It was designed for content based recommendation. Representation and aggregation methods were empirically tested on movie dataset. In [55] quality of recommendation was improved by maintaining recall with high precision. The models developed for managing the dynamics of recommender system proved to be efficient.

After 2011, Fuzzy logic based recommender systems found new applications in various domains. For instance, in [56], fuzzy logic based recommender system was used for tag-clouds. The fuzzy theory worked on the process of finding the strength of tag-item connections. It is observed that there is an increase in the growth of fuzzy logic based recommender system using collaborative filtering and hybrid systems [26, 34, 36, 41, 43]. Now multi-criteria collaborative filtering has also been used in tourism domain [57] to understand user preferences in multiple aspects. This also deals with the dynamic nature of recommender system. The future of fuzzy logic based recommender system is witnessing attention in both collaborative and hybrid systems with soft computing techniques as well. Collaborative filtering is being studied with fuzzy theory to understand novel video recommendation [58],

with swarm intelligence techniques [59], for interior design drawing [60], social network [61].

5 Conclusion

The introduction of fuzzy logic based recommender system in 2003 by Yager, has attracted attention of many researchers to advance into this field to find new methods to deal with uncertainty in data, user preference and user demographics. This uncertainty is inherent in human behaviour and a careful analysis and understanding of the basic functioning of the recommender system definitely helps in predicting accurate recommendations for the user. This chapter, described the recommender system as an information filtering system with a discussion on the analytical framework of the recommender system. It enlightened the reader about the theory and applications of recommender system. It, also, qualitatively and quantitatively analysed the growth of fuzzy logic based recommender system and its applications from movies, tourism, stocks to social network. This chapter clearly explained fuzzy logic based recommender system (reclusive, collaborative filtering and hybrid systems) and its applications in various domains.

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